The documentation and process conversion measures necessary to comply with this revision shall be completed by 8 July 1994

INCH POUND

MIL-S-19500/421D
8 April 1994
SUPERSEDING
MIL-S-19500/421C(USAF)
13 March 1992

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, UNITIZED, DUAL TRANSISTOR, NPN/PNP, COMPLEMENTARY, SILICON
TYPES 2N3838, 2N4854, AND 2N4854U
JANTX AND JANTXV

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for unitized, dual transistors which contain a pair of electrically isolated complementary NPN and PNP silicon triode transistors in one package. Two levels of product assurance are provided for each device type as specified in MIL-S-19500.

1.2 Polarity designation. Voltages and currents of limits and test conditions shown herein apply to the NPN triode. For the PNP triode, the values are the same, but the polarity designations shall be the opposite.

1.3 Physical dimensions. See 3.3 (6 lead flatpak, can, and surface mount).

1.4 Maximum ratings.

P_T at T_A = +25°C				P_T at T_C = +25°C 1/				V_{CBO}
2N3838 2/		2N4854 3/		2N3838 4/		2N4854 5/		
One Transistor	Total Device	One Transistor	Total Device	One Transistor	Total Device	One Transistor	Total Device	
W	W	W	W	W	W	W	W	V dc
0.25	0.35	0.30	0.60	0.7	1.4	1.0	2.0	60

V_{EBO}	V_{CEO}	V_{1C-2C}	Lead to case voltage	I_C	T_J	T_{STG}
V dc	V dc	V dc	V dc	mA dc	°C	°C
5	40	±120	±120	600	200	-55 to +200

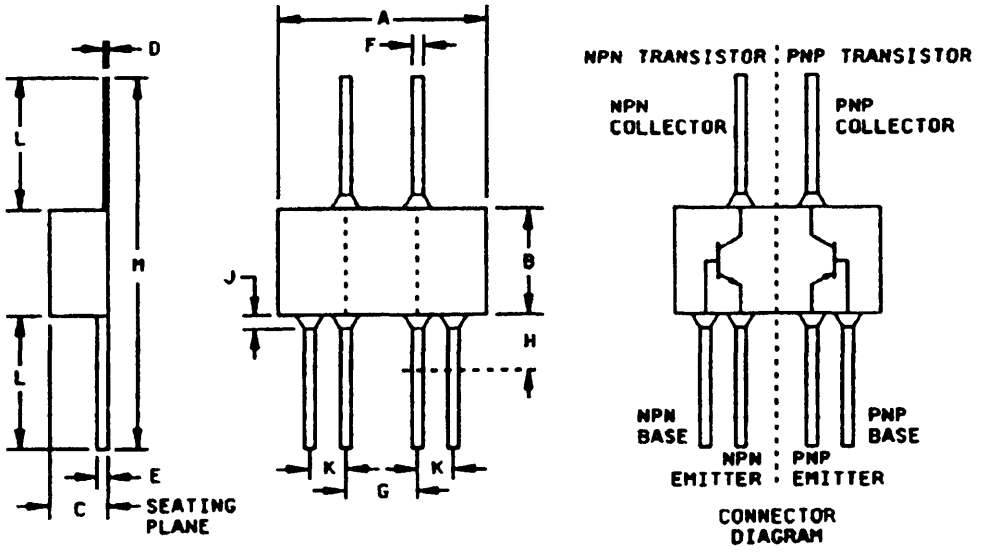
1/ T_C rating do not apply to surface mount devices (2N4854U).
2/ For T_A > +25°C, derate linearly 1.43 mW/°C one transistor, 2.00 mW/°C both transistors.
3/ For T_A > +25°C, derate linearly 1.71 mW/°C one transistor, 3.43 mW/°C both transistors.
4/ For T_C > +25°C, derate linearly 4.0 mW/°C one transistor, 8.0 mW/°C both transistors.
5/ For T_C > +25°C, derate linearly 5.71 mW/°C one transistor, 11.43 mW/°C both transistors.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Electronics Supply Center, ATTN: DESC-ELDT, 1507 Wilmington Pike, Dayton, OH 45444-5765, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A
FSC 5961

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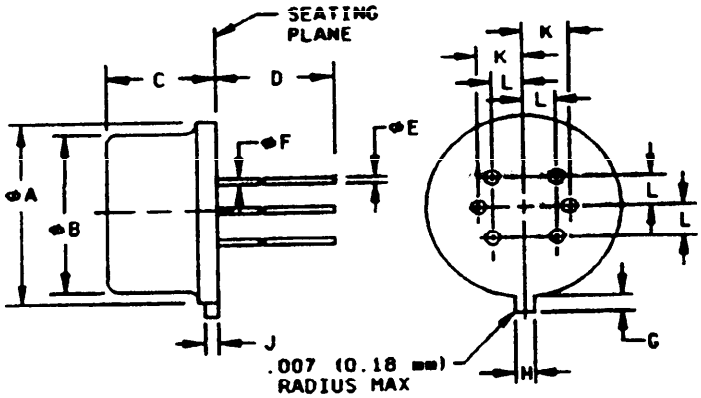
Ltr	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
A	.240	.290	6.10	7.37	
B	.115	.160	2.92	4.06	
C	.030	.080	0.76	2.03	
D	.003	.006	0.08	0.15	4
E	.005	.035	0.13	0.89	
F	.010	.019	0.25	0.48	4, 6

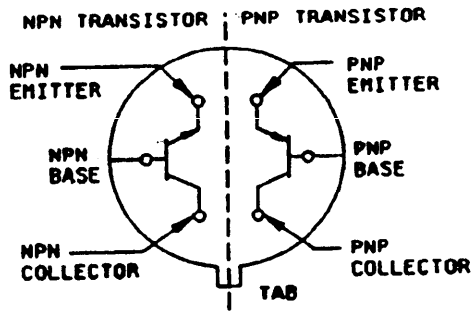
Ltr	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
G	.100 TP		2.54 TP		6, 7
H		.050		1.27	
J		.015		0.38	5
K	.050 TP		1.27 TP		6, 7
L	.070	.250	1.78	6.35	3, 4
M	.260	.650	6.60	16.51	

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Maximum limit of this dimension does not apply to device supplied in a carrier.
4. All six leads.
5. Lead dimensions are uncontrolled in this zone.
6. Dimensions "F", "G", and "K" to be measured in zone "H".
7. Leads within .005 inch (0.13 mm) total of true position (TP) at "H" with maximum material condition.

FIGURE 1. Physical dimensions of transistor type 2N3838 (all quality levels).

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CONNECTION DIAGRAM

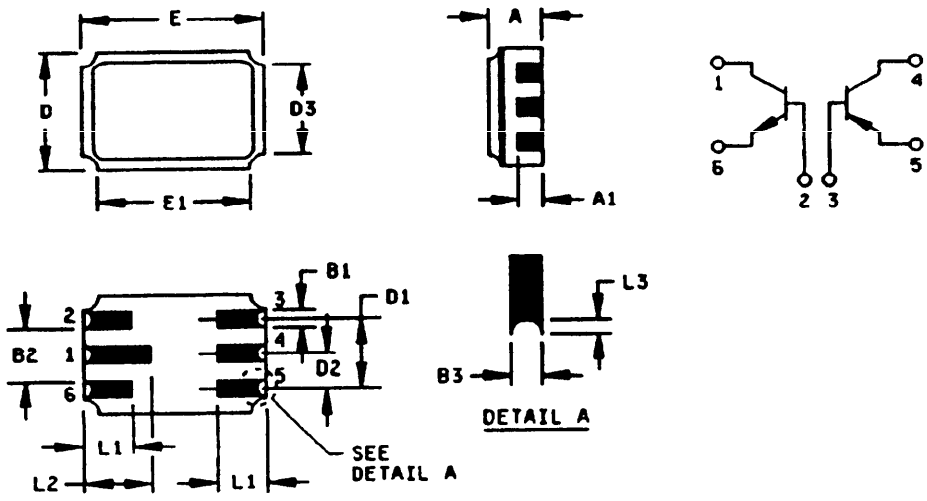
Ltr	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
ΦA	.335	.370	8.51	9.40	
ΦB	.305	.335	7.75	8.51	
C	.140	.260	3.56	6.60	
D	.500	1.750	12.70	44.45	7
ΦE	.016	.021	0.41	0.53	3, 7
ΦF	.016	.019	0.41	0.48	4, 7

Ltr	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
G	.029	.045	0.74	1.14	6
H	.028	.034	0.71	0.86	
J	.009	.125	0.23	3.18	
K	.1000 Nom.		2.540 Nom.		5
L	.0707 Nom.		1.796 Nom.		5

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Measured in the zone beyond .250 inch (6.35 mm) from the seating plane.
4. Measured in the zone .050 inch (1.27 mm) and .250 inch (6.35 mm) from the seating plane.
5. When measured in a gauging plane .054 +.001, -.000 inch (1.37 +0.03, -0.00 mm) below the seating plane of the transistor, maximum diameter leads shall be within .007 inch (0.18 mm) of their true location relative to a maximum width tab. Smaller diameter leads shall fall within the outline of the maximum diameter lead tolerance.
6. Measured from the maximum diameter of the actual device.
7. All six leads.

FIGURE 2. Physical dimensions of transistor type 2N4854 (all quality levels).

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Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.066	.080	1.68	2.03
A1	.026	.039	0.66	0.99
B1	.022	.028	0.56	0.71
B2	.072 Ref		1.83 Ref	
B3	.006	.022	0.15	0.56
D	.165	.175	4.19	4.45
D1	.095	.105	2.41	2.67

Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
D2	.045	.055	1.14	1.40
D3		.175		4.45
E	.240	.250	6.10	6.35
E1		.250		6.35
L1	.060	.070	1.52	1.78
L2	.082	.098	2.08	2.49
L3	.003	.007	0.08	0.18

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The coplanarity deviation of all terminal contact points, as defined by the device seating plane, shall not exceed .006 inch (0.15 mm) for solder dipped leadless chip carriers.

FIGURE 3. Physical dimensions of transistor type 2N4854U.

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1.5 Primary electrical matching characteristics of each individual section. Characteristics apply to all case outlines.

	h_{FE5} 1/ V_{CE} = 10 V dc I_C = 150 mA dc	h_{FE3} 1/ V_{CE} = 10 V dc I_C = 10 mA dc	h_{FE2} V_{CE} = 10 V dc I_C = 100 µA dc	$V_{CE(sat)}$ 1/ I_B = 15 mA dc I_C = 150 mA dc	$V_{BE(sat)}$ 1/ I_B = 15 mA dc I_C = 150 mA dc
				V dc	V dc
MIN	100	75	35		0.80
MAX	300			0.4	1.25

	$\|h_{fe}\|$ V_{CE} = 10 V dc I_C = 20 mA dc f = 100 MHz	C_{obo} V_{CB} = 10 V dc I_E = 0 100 kHz ≤ f ≤ 1 MHz	V_{CEO} 2/ $I_{C(on)}$ = 600 mA dc $I_{B(on)}$ = 120 mA dc $I_{B(off)}$ = 0 See figure 7	$t_{(on)}$ See figure 4	$t_{(off)}$ See figure 5	$t_{on} + t_{off}$ See figure 6
		pF	V dc	ns	ns	ns
MIN	2		40			
MAX	8	8		45	300	18

1/ Pulsed (see 4.5.1).
2/ See 4.5.2.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Defense Printing Service Detachment Office, Building 4D (Customer Service), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein the text of this document take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Associated Detail specification. The individual item requirements shall be in accordance with MIL-S-19500 and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500 and as follows:

V_{1C-2C} . . . . . Voltage difference between the collector of transistor one and that of transistor two.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500 and on figures 1, 2, and 3 herein.

3.3.1 Lead finish. Lead finish shall be solderable as defined in Mil-S-19500, MIL-STD-750, and herein.

3.4 Marking. Marking shall be in accordance with MIL-S-19500.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500 and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500.

4.3 Screening (JANTX and JANTXV levels). Screening shall be in accordance with tab;e II of MIL-S-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table II of MIL-S-19500)	Measurement
	JANTX and JANTXV levels
11	I_{CBO1}, h_{FE5}
12	See 4.3.1
13	Subgroup 2 of table I herein; ΔI_{CBO1} = 100% or 25 nA, whichever is greater, for the 2N3838; ΔI_{CBO1} = 100% or 5 nA whichever is greater, for the 2N4854. Δh_{FE5} = ±15%

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows: T_A = room ambient as defined in the general requirements of MIL-STD-750, (see 4.5); P_T = 1.0 W;

2N3838: P_T = 175 mW each transistor (350 mw total device).
2N4854: P_T = 300 mW each transistor (600 mW total device).
2N4854U: P_T = 300 mW each transistor (600 mW total device).

NOTE: No heat sink or forced air cooling on the devices shall be permitted.

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4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500, and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVa (JANS) and table IVb (JANTX and JANTXV) of MIL-S-19500. Electrical measurements (end-points) shall be in accordance table I, group A, subgroup 2 herein.

4.4.2.1 Group B inspection, table IVb (JANTX and JANTXV) of MIL-S-19500.

Subgroup	Method	Condition
B3	1027	V_{CB}= 30 V dc; 2N3838: P_T = 175 mW each transistor (350 mW total device); 2N4854, 2N4854U: P_T = 300 mW each transistor (600 mW total device) at T_A = +30°C ±5°C. No heat sink or forced-air cooling on the devices shall be permitted.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500, and as follows. Electrical measurements (end-points) shall be in accordance table I, group A, subgroup 2 herein.

4.4.3.1 Group C inspection, table V of MIL-S-19500.

Subgroup	Method	Condition
C2	2036	Test condition E; 2N3838: 3 oz. weight, 3 bends of 45°; 2N4854: 8 oz. weight, 3 bends of 90°(Does not apply to 2N4854U).
C6	1026	1,000 hours at V_{CB}= 30 V dc; 2N4854, 2N4854U: P_T = 300 mW (one section); P_T = 600 mW (both sections); 2N3838: P_T = 175 mW each transistor (350 mW total device); at T_A = 30°C ±5°C. No heat sink or forced-air cooling on the device shall be permitted.

4.5 Method of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Testing of units. All specified electrical tests, including end-point tests, shall be performed equally on both sections of the transistor types covered herein, except where the electrical characteristic being evaluated applies to the transistor as a device entity.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Breakdown voltage, collector to base	3001	Bias condition D; I_C = 10 µA dc	$V_{(BR)CBO}$	60		V dc
Breakdown voltage, collector to emitter	3011	Bias condition D; I_C = 10 mA dc; pulsed (see 4.5.1)	$V_{(BR)CEO}$	40		V dc
Breakdown voltage, emitter to base	3026	Bias condition D; I_E = 10 µA dc	$V_{(BR)EBO}$	5		V dc
Collector to base cutoff current	3036	Bias condition D; V_{CB} = 50 V dc	I_{CBO1}			
2N3838					50	nA dc
2N4854					10	nA dc
2N4854U					10	nA dc
Emitter to base cutoff current	3061	Bias condition D; V_{EB} = 3 V dc	I_{EBO}		10	nA dc
Forward-current transfer ratio	3076	V_{CE} = 1 V dc; I_C = 150 mA dc; pulsed (see 4.5.1)	h_{FE1}	50		
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 100 µA dc	h_{FE2}	35		
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 1.0 mA dc	h_{FE3}	50		
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 10 mA dc; pulsed (see 4.5.1)	h_{FE4}	75		
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 150 mA dc; pulsed (see 4.5.1)	h_{FE5}	100	300	
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 300 mA dc; pulsed (see 4.5.1)	h_{FE6}	35		
Base emitter saturation voltage	3066	Test condition A; I_B = 15 mA dc; I_C = 150 mA dc; pulsed (see 4.5.1)	$V_{BE(sat)}$	0.80	1.25	V dc

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 2 - Continued						
Collector to emitter saturation voltage	3071	I_B = 15 mA dc; I_C = 150 mA dc; pulsed (see 4.5.1)	$V_{CE(sat)}$		0.40	V dc
Subgroup 3						
High temperature operation:		T_A = +150°C				
Collector to base cutoff current	3036	Bias condition D; V_{CB} = 50 V dc	I_{CBO2}		10	µA dc
Low temperature operation:		T_A = -55°C				
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 10 mA dc	h_{FE7}	12		
Subgroup 4						
Small-signal short-circuit forward current transfer ratio	3206	V_{CE} = 10 V dc; I_C = 1 mA dc; f = 1 kHz	h_{fe}	60	300	
Small-signal common emitter input impedance	3201	V_{CE} = 10 V dc; I_C = 1 mA dc; f = 1 kHz	h_{ie}	1.5	9	kΩ
Small-signal common emitter output admittance	3216	V_{CE} = 10 V dc; I_C = 1 mA dc; f = 1 kHz	h_{oe}		50	µmho
Magnitude of small-signal short-circuit forward current transfer ratio	3306	V_{CE} = 10 V dc; I_C = 20 mA dc; f = 100 MHz	$\|h_{fe}\|$	2	8	
Open circuit output capacitance	3236	V_{CB} = 10 V dc; I_E = 0; 100 kHz ≤ f ≤ 1 MHz	C_{obo}		8	pF
Noise figure	3246	V_{CE} = 10 V dc; I_C = 100 µA dc; f = 1 kHz; R_G = 1 kΩ	NF		8	dB

See footnote at end of table.

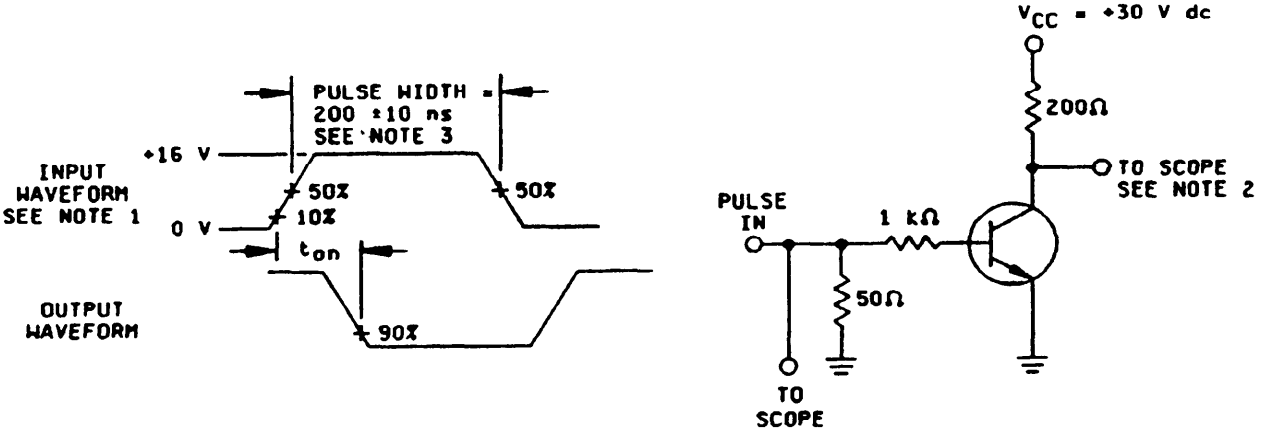
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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 4 - Continued						
Turn-on time		See figure 4	t_{on}		45	ns
Turn-off time		See figure 5	t_{off}		300	ns
Pulse response		See figure 6	$t_{on} + t_{off}$		18	ns
Collector emitter nonlatching voltage		See figure 7	V_{CEO}	40		V dc
Subgroups 5, 6, and 7						
Not applicable						

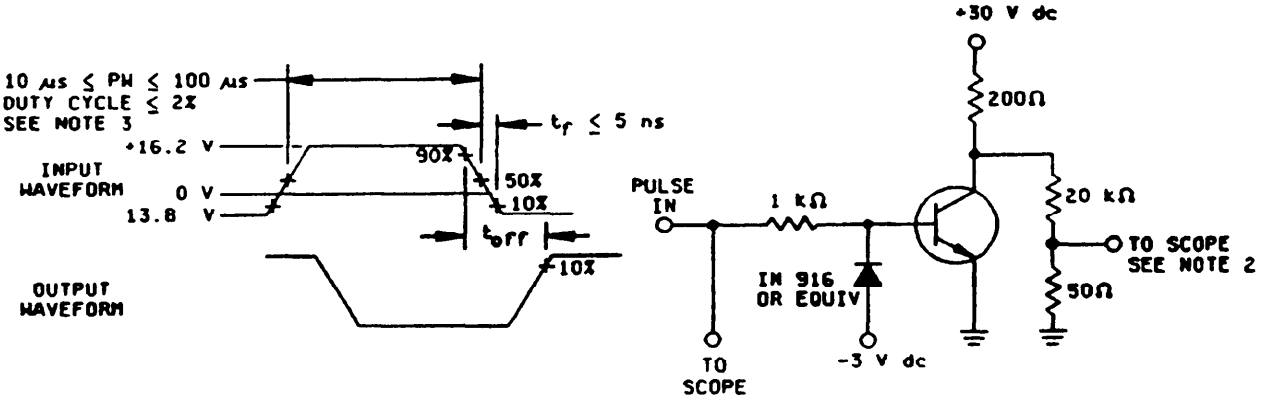
1/ For sampling plan see MIL-S-19500.

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NOTES:
1. The rise time (t_r) of the applied pulse shall be ≤ 2.0 ns, duty cycle ≤ 2 percent, and the generator source impedance shall be 50 Ω.
2. Sampling oscilloscope: Z_{IN} ≥ 100 kΩ, C_{IN} ≤ 12 pF, rise time ≤ .2 ns.
3. The signs, polarity symbols, and waveforms shown are for the NPN triode; the signs and polarity symbols are reversed for the PNP triode (see 1.2 herein).

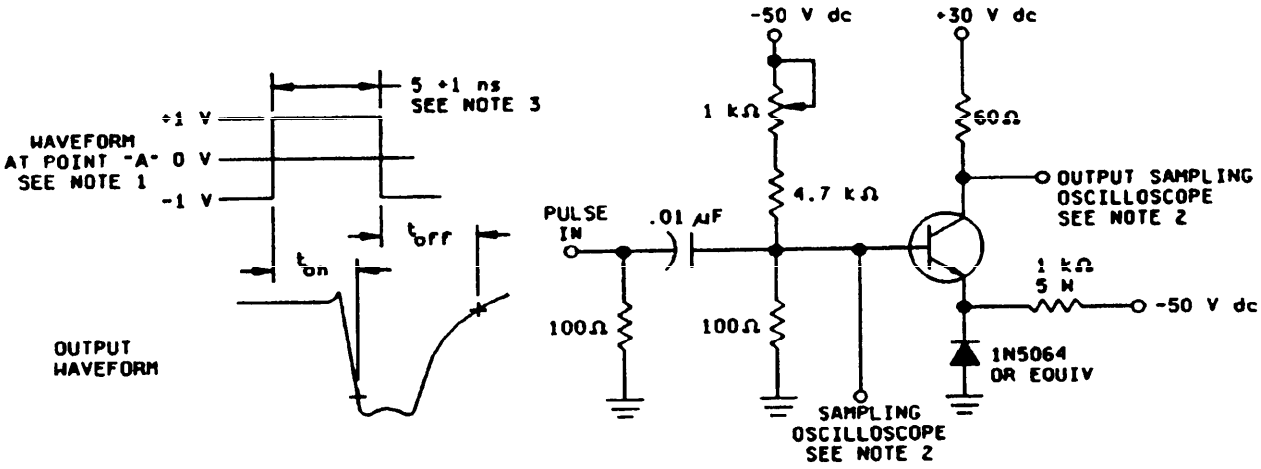
FIGURE 4. Saturated turn-on switching time test circuit.



NOTES:
1. The rise time (t_r) of the applied pulse shall be ≤ 2.0 ns, duty cycle ≤ 2 percent, and the generator source impedance shall be 50 ohms.
2. Sampling oscilloscope: Z_{IN} ≥ 100 kΩ, C_{IN} ≤ 12 pF, rise time ≤ .2 ns.
3. The signs, polarity symbols, and waveforms shown are for the NPN triode; the signs and polarity symbols are reversed for the PNP triode (see 1.2 herein).

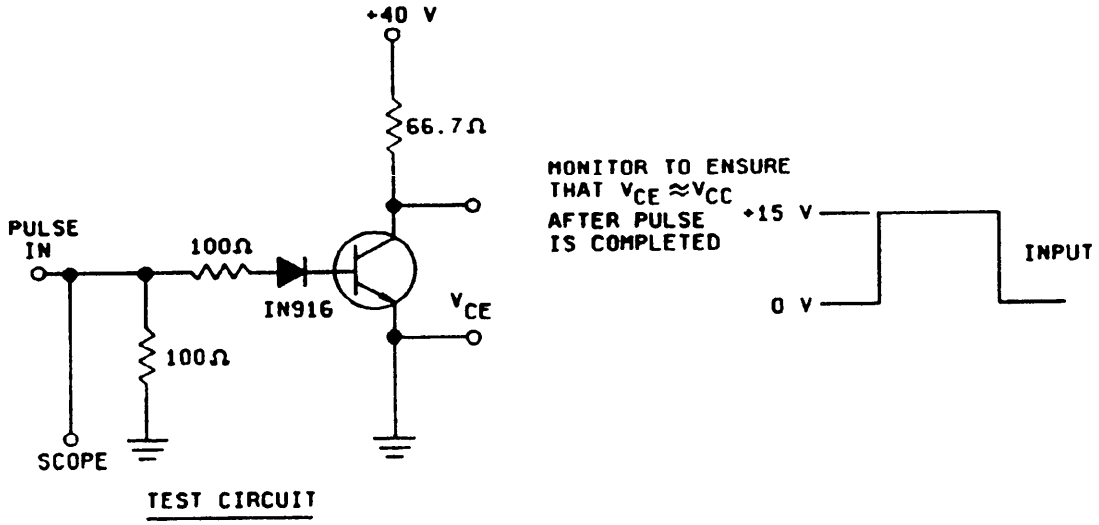
FIGURE 5. Saturated turn-off switching time test circuit.

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NOTES:
1. The rise time (t_r) of the applied pulse shall be ≤ 2.0 ns, duty cycle ≤ 2 percent, and the generator source impedance shall be 50 Ω.
2. Sampling oscilloscope: Z_{IN} ≥ 100 kΩ, C_{IN} ≤ 12 pF, rise time ≤ .2 ns.
3. The signs, polarity symbols, and waveforms shown are for the NPN triode; the signs and polarity symbols are reversed for the PNP triode (see 1.2 herein).

FIGURE 6. Nonsaturated switching time test circuit.



NOTE: The input waveform has the following characteristics: PW ≤ 10 μs, duty cycle ≤ 2 percent.

FIGURE 7. Collector emitter nonlatching voltage test circuit.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Issue of DODISS to be cited in the solicitation.

b. Lead finish (see 3.3.1).

c. Type designation and product assurance level.

6.3 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodian:
Air Force - 17

Preparing activity:
DLA - ES

(Project 5961-F116)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-S-19500/421D	2. DOCUMENT DATE (YYMMDD) 94/04/08

3. DOCUMENT TITLE SEMICONDUCTOR DEVICE, UNITIZED, DUAL TRANSISTOR, NPN/PNP, COMPLEMENTARY, SILICON TYPES 2N3838, 2N4854 AND 2N4854U JANTX AND JANTXV

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, first, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Alan Barone	b. TELEPHONE (Include Area Code) (1) Commercial 513-296-6048 (2) AUTOVON 986-6048
c. ADDRESS (Include Zip Code) Defense Electronics Supply Center ATTN: DESC-ELDT Dayton, Ohio 45444-5765	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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